Spatial Analysis of Australian Bushfire Intensity: Black Saturday vs Black Summer

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Introduction

This report examines and compares two of the most catastrophic bushfire seasons in Australian history: the 2008-2009 Black Saturday fires and 2019-2020 Black Summer bushfires. The first dataset captures the period where the Black Saturday fires devastated large portions of Queensland and Victoria. The second dataset encompasses the Black Summer fires, which burned over 18 million hectares nationwide and caused severe ecological and environmental damage.

Objective

This study employs geospatial data analysis to compare two significant bushfire events using Fire Radiative Power (FRP) data, which will be visualised and analysed graphically. The primary objectives are to find differences and similarities in fire extent, spatial distribution, and intensity.
These findings may provide critical insights into the influence of the evolving climate, land management practices, and vegetation changes on scale, behavior and resultant impact of Australian bushfires over a ten-year period.

Methodology

The analysis was conducted using spatial bushfire data for the 2008-2009 and 2019-2020 fire seasons. Both datasets were imported into QGIS for processing and visualisation.

1. **Data Sources**
   - **Bushfire Data (Point Features):**
     Two separate **CSV** datasets of Australian bushfires were obtained from **NASA's Fire Information for Resource Management System (FIRMS)**, corresponding to the periods (**1 December 2008 to 28 February 2009**) & (**1 September 2019 to 31 March 2020**), respectively. The datasets included key attributes such as **latitude, longitude, and Fire Radiative Power (FRP)**.

- **Base Map Layer (Raster Imagery):**
  A high resolution **ESRI Satellite** imagery layer was used to provide essential geographic context for visualising the fire locations. This layer offers a detailed visual reference of land cover and topography, aiding in the interpretation of fire locations relative to built-up areas, vegetation, and major geographic features. The layer was accessed via the Web QuickMapServices plugin in QGIS.

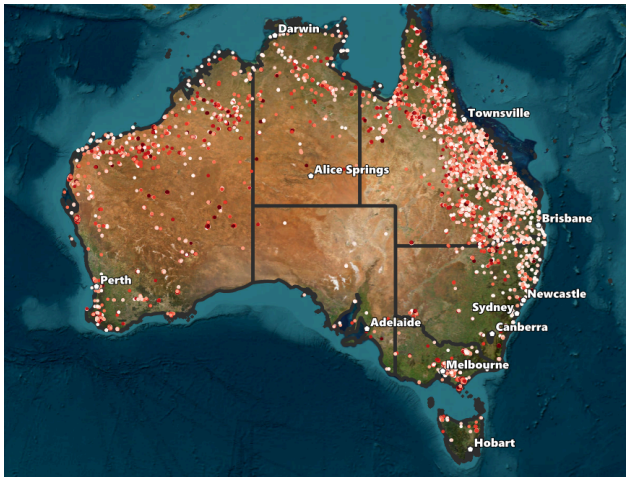
- **States Boundaries (Polygon Layer):**
  The administrative boundaries for Australian states and territories were included as a polygon layer to indicate the areas affected by the fires The layer source is the **GADM (Global Administrative Boundaries)** dataset.

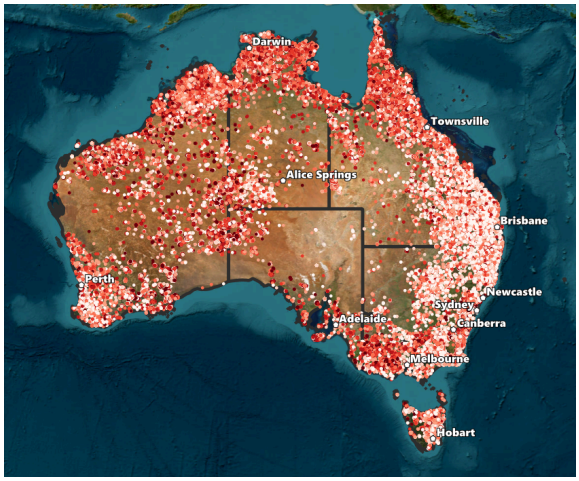
- **Major Cities (Point Features):**
  Australian major cities were included as reference points to provide geographic orientation for the viewer. This point feature dataset was sourced from the **Natural Earth Data** project (Cultural Section - **10m_populated_place_simple shape file**). The data was filtered in QGIS using the expression: `("ADM0NAME" = 'Australia' AND "SCALERANK" <= 3)` to include only the largest Australian cities.

2. Symbology and Visualisation

The bushfire points were symbolized according to their **fire intensity** using graduated coloured circles (or “pointer markers”). A **white-to-dark-red** colour ramp was applied, where lighter points represent lower intensity (low FRP) and darker points represent high intensity (high FRP). The classification method used was **Natural Breaks (Jenks)** to ensure the class intervals accurately reflect the skewed distribution of the Fire Radiative Power data.

Comparative Analysis





Left Picture: Black Saturday Fires (2008-2009)

The 2008-2009 map shows mostly mild and scattered bushfire activity. Coastal Queensland contains several clusters, especially around Brisbane and Townsville, along with smaller concentrations near Melbourne and along the southern coast of Western Australia. Northern WA also shows scattered activity, with some high-FRP incidents occurring in remote inland reserves. Central Australia displays moderate but noticeably lighter activity overall. Compared to the later dataset, the fires appear more localised, with large regions—especially the east coast, Tasmania, and the Top End—remaining relatively low in density.

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Right Picture: Black Summer Fires (2019-2020)

The 2019-2020 map is significantly more saturated, showing widespread fire activity across nearly all regions of Australia. Northern coastal areas, including the Top End, Arnhem Land, and the Gulf regions, show heavy high-FRP clustering. Darwin sits within a broad zone of intense burning rather than an isolated hotspot. A continuous band of moderate to severe fire activity stretches along the entire eastern coastline from Townsville through Brisbane and Sydney down to Melbourne, with New South Wales experiencing a dramatic increase in density compared to 2008-2009. Victoria shows major high-FRP clusters, especially in its western forests. Inland Queensland and central Australia also display far denser activity than the earlier period. Tasmania shifts from sparse

activity high-density low-moderate FRP fires across much of the island. Western Australia shows a similar expansion, with larger and more intense clusters in the north and interior regions.

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Direct Comparison (Key Differences)

The visual contrast between the two maps is substantial.

- ➢ Fire density in 2019-2020 is dramatically higher nationwide.
- ➢ Regions that were mild or sparsely affected in 2008-2009—New South Wales, inland Queensland, the Top End, Victoria, Tasmania—show major increases in both frequency and severity.
- ➢ Central Australia and northern WA also shift from scattered patterns to dense, widespread activity.
- ➢ Overall, the 2019-2020 period demonstrates a national-scale escalation, with nearly the entire continent exhibiting significantly more fire activity compared to the localised clusters of 2009-2009.

Discussion

The spatial patterns in both datasets indicate that bushfire activity is heavily influenced by vegetation availability rather than by simple coastal-inland differences. Areas with dense fuels—such as forests, woodlands, and large nature reserves—consistently appear as fire hotspots in both time periods. This explains why fires occur across such a wide range of environments, including coastal regions, inland savannas, and remote reserves. The maps show that vegetation type and fuel load are far more predictive of fire occurrence than proximity to the coast.

A key factor underlying both datasets is the timing: the recorded months coincide with the Australian summer, when temperatures peak and vegetation dries out. High heat, low humidity, and strong winds sharply increase fire ignition and spread potential. This seasonal alignment means that regions with adequate fuel loads become far more vulnerable during these months, regardless of whether they are coastal, inland, tropical or temperate.

Seemingly unexpected clusters, such as those near Alice Springs, can be explained by the presence of fire-prone vegetation like spinifex grasslands and shrublands. Although central Australia is generally perceived as arid, these systems accumulate enough dry fuel to support large burns when seasonal conditions are favourable. This demonstrates that “dry” does not necessarily mean “low fire activity”; fire frequency depends on the combination of fuel availability, ignition sources, and seasonal climate patterns.

Limitations

1. The satellite data shows fire detections as points, which can overlap in high-density regions and make it difficult to see **individual** fire events.

2. Fire Radiative Power (FRP) indicates **intensity** but does not show total burned area or duration of a fire.

3. Cloud cover, smoke, or sensor limitations may cause missing or delayed detections in some regions.

4. The maps represent specific seasonal periods rather than full-year patterns, so lower-season fires are not captured.

5. Regional vegetation types are **generalised** and may not perfectly reflect local fuel conditions.

Conclusion

The comparison shows that bushfire activity in 2019-2020 was significantly more widespread and intense than in 2008-2009. Areas with substantial vegetation—whether coastal, inland or remote—consistently produced the highest fire concentrations, emphasising the role of fuel load over simple geography location. The timing of both datasets during the summer months further contributed to increased fire activity due to hotter, drier conditions. Overall, the maps demonstrate how vegetation distribution and seasonal climate combine to shape Australia's bushfire patterns at a national scale.

Glossary of Regional Terms

Top End
Refers to the northernmost region of the Northern Territory, including areas around Darwin. It is characterised by tropical climate, monsoon forests, and extensive savanna vegetation.

Arnhem Land
A large, remote region in the northeast of the Northern Territory. It is known for rugged terrain, dense vegetation, and a long history of seasonal fire activity.

Gulf Region (Gulf Country)
The area surrounding the Gulf of Carpinteria in northern Australia. It includes parts of Queensland and the Northern Territory, dominated by grasslands and savanna woodland.

Eastern Seaboard
The coastal stretch running from Queensland through New South Wales down to Victoria. It includes major population centres such as Brisbane, Sydney, and Melbourne.

Interior / Inland Australia
Refers to central and remote regions away from the coast, including deserts, rangelands, and sparsely populated areas such as central WA, NT, and SA.

FRP (Fire Radiative Power)
A measurement of fire intensity based on the heat energy released. Higher FRP values typically indicate more intense or actively burning fires.